



SECURITY
INTELLIGENCE
PLATFORM

Infosys Springboard
Virtual Internship 7.0

Batch 2 | Data Visualization

Advanced Threat Incident Analytics & Security Visualization Framework

"Intelligent Security. Actionable Insights. Safer Tomorrow."

7

Analytics Modules

4

Delivery Milestones

8

Week Rollout Plan

? Project Statement

⚠ THE CHALLENGE

Organizations responsible for public safety, critical infrastructure, enterprise security, and emergency management generate large volumes of incident, threat, operational, environmental, and situational data.

This data is scattered across multiple systems — making it difficult to:

- Identify emerging threats on time
- Understand incident patterns across locations
- Evaluate response effectiveness vs SLA targets
- Proactively mitigate future risks
- Make fast and informed decisions

✅ THE FRAMEWORK

The Advanced Threat Incident Analytics and Security Visualization Framework consolidates threat incidents, emergency events, environmental risk indicators, and operational response records into a unified analytics environment.

Enabling:

- Detect threat hotspots
- Monitor incident trends
- Optimize response strategies
- Assess operational performance
- Forecast emerging risk zones

"Too much data — but no clear picture of what is happening."



About Our Dataset



1,00,000

Total Records



15

Total Columns



10

Countries



8

Industries



8

Attack Types



All 15 Columns & Their Purpose

♦ **attack_type**

Type of attack — Phishing, DDoS, Ransomware etc.

♦ **target_system**

System attacked — Cloud, IoT, Database, Email Server

♦ **outcome**

Attack result — Success or Failure

♦ **timestamp**

When the attack happened — date & time

♦ **data_compromised_GB**

How much data was stolen in GB

♦ **attack_severity**

Danger level of attack — score from 1 to 10

♦ **location**

Which country was attacked

♦ **industry**

Sector attacked — Finance, Energy, Healthcare etc.

♦ **response_time_min**

How fast team responded in minutes

♦ **mitigation_method**

How attack was stopped — Patch, Quarantine etc.

♦ **security_tools_used**

Tool used — Firewall, MFA, Antivirus etc.

♦ **user_role**

Who was targeted — Admin, Employee, External User

♦ **attack_duration_min**

How long the attack lasted in minutes

♦ **attacker_ip**

IP address of attacker

♦ **target_ip**

IP address of victim



Data Cleaning — What We Did & Why

Missing Values

Before: 8 cells

After: 0 cells

Duplicate Rows

Before: 3 rows

After: 0 rows

Invalid Values

Before: 4 values

After: 0 values

Data Quality

Before: 68%

After: 100%



6 Steps We Followed:



Step 1

Missing Values

Filled empty cells with "Unknown" or "None" using fillna()



Step 2

Invalid Values

Fixed attack_severity > 10, negative response time, negative data_GB



Step 3

Duplicate Rows

Removed 3 exact duplicate rows using drop_duplicates()



Step 4

Inconsistent Text

Standardized case using str.capitalize() and str.strip()



Step 5

Wrong Timestamp

Converted all dates to YYYY-MM-DD HH:MM:SS format using pd.to_datetime()



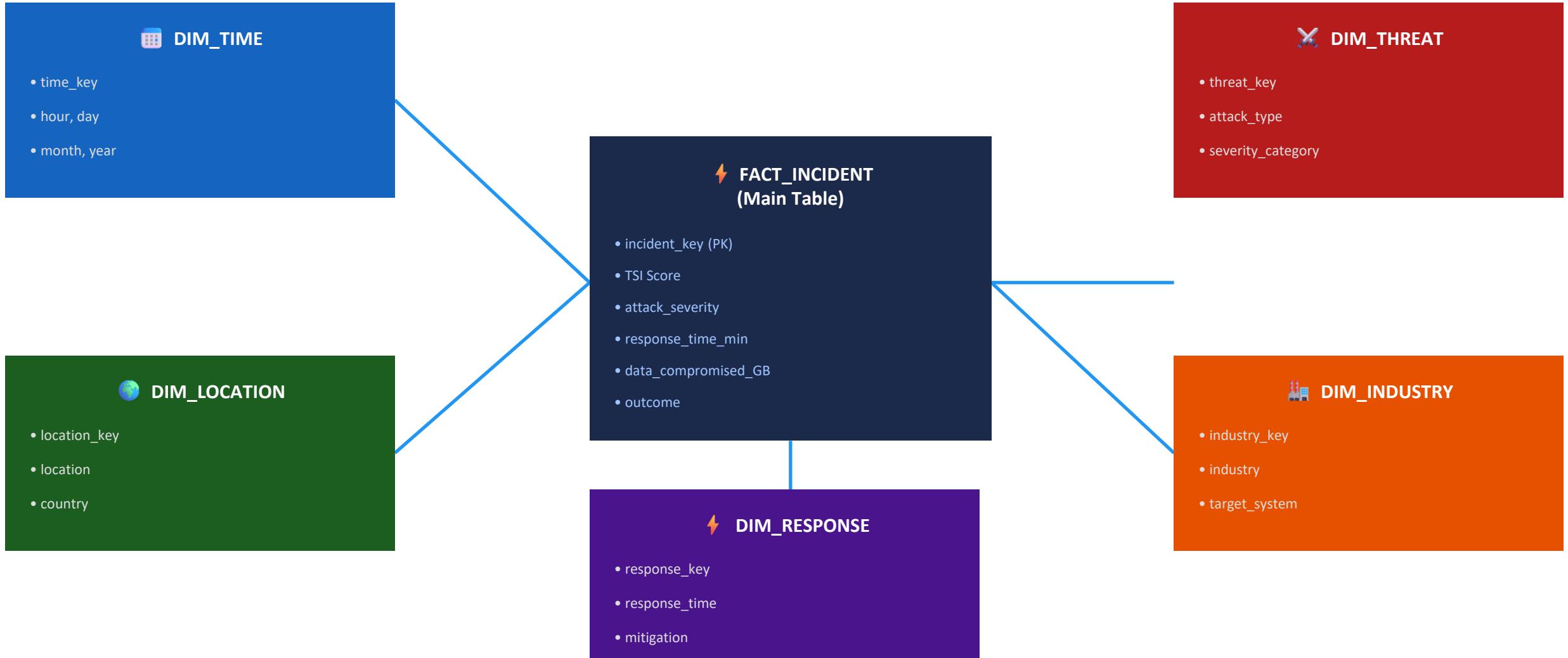
Step 6

Data Type Mismatch

Converted columns to correct int, float, and datetime types

★ Star Schema — How We Organized Data

We divided our dataset into 1 Fact Table (center) and 5 Dimension Tables around it — like a star!



All tables connected through keys | 1 Fact Table + 5 Dimension Tables = Star Schema ★

TSI Score — Threat Severity Index

TSI tells us how dangerous each attack is — combining 3 factors into one score.

$$\text{TSI} = (\text{Attack Severity} \times 0.4) + (\text{Data Compromised} / 100 \times 10 \times 0.3) + (\text{Response Time} / 180 \times 10 \times 0.3)$$



Attack Severity

Weight: 40%

How dangerous the attack is

Example:
Severity = 7
 $7 \times 0.4 = 2.8$



Data Compromised

Weight: 30%

How much damage was done

Example:
Data = 65 GB
 $(65/100 \times 10) \times 0.3 = 1.95$



Response Time

Weight: 30%

How late the team responded

Example:
Time = 90 min
 $(90/180 \times 10) \times 0.3 = 1.5$

TSI Score Range:

0–3 Low

3–5 Medium

5–7 High

7–10 Critical

Our Dataset TSI Score: 5.21 — HIGH THREAT 



8-Week Delivery Roadmap — 4 Milestones

Milestone 1

Weeks 1–2

Data Integration & Incident Modeling

- Collect & clean dataset (1 lakh records)
- Build Python ETL pipeline
- Design Star Schema (1 fact + 5 dim)
- Calculate TSI Score baseline

✓ Done

Milestone 2

Weeks 3–4

Threat Intelligence & Temporal Analytics

- Build TSI Gauge dashboard
- Classify 8 attack types
- Monthly trend analysis
- Security tools usage analysis

✓ Done

Milestone 3

Weeks 5–6

Geospatial Intelligence & Response Analytics

- World threat hotspot map
- Country-wise response time bar
- Industry × Location heatmap
- Response performance metrics

✓ Done

Milestone 4

Weeks 7–8

Risk Intelligence & Executive Analytics

- Security Risk Scorecard
- Industry risk analysis
- Platform risk gauge
- Executive command center

✓ Done



Dashboard Page 1 — Overview

Milestone 1 — Weeks 1–2



Total Incidents

1,00,000

All recorded attacks



Avg TSI Score

5.21

High Threat Level



Avg Response Time

90 Min

Target: under 60 min



Data Compromised

5.01M GB

Total across all attacks



Bar Chart — Incidents by Attack Type



Horizontal bars showing count of each attack type



Why: Best for comparing 8 categories — see which attack is most common



Donut Chart — Attack Outcome



Shows Success vs Failure split of all 1 lakh attacks



Why: Pie/donut best for showing proportion — 50% success vs 50% failure



Donut Chart — Severity Distribution



Shows Critical, High, Medium, Low % split



Why: Shows distribution — Critical 12.5%, Medium 37.63%



Column Chart — Most Targeted Industries



8 industries compared by incident count



Why: Column chart best for multiple categories — Government highest at 12,645



Dashboard Page 2 — Threat Intelligence

Milestone 2 — Weeks 3–4

⚡ Overall TSI Score

5.21

HIGH Threat Level

🔹 Critical Incidents

12,499

Ransomware only

📈 High Severity Count

25,024

Zero-Day & Malware

🧠 Avg Severity Score

5.50

Out of 10 max

⚡ Gauge Chart — Threat Severity Index

📌 Semicircle gauge showing TSI 5.21 out of 10

✅ Why: Gauge best for showing a score within a range — Red zone = High threat

📅 Bar Chart — Monthly Incident Trend

📌 12 months with incident count — March highest 8.6K

✅ Why: Bar chart shows monthly comparison — multi-color highlights peaks

🍷 Donut Chart — Incident Categories

📌 8 attack types each at ~12.5% distribution

✅ Why: Donut shows proportional split — no single type dominates

🔧 Bar Chart — Security Tools Usage

📌 8 tools ranked — Endpoint Detection leads at 145,231

✅ Why: Bar chart compares tool usage — helps prioritize security investments



Dashboard Page 3 — Geospatial & Response

Milestone 3 — Weeks 5–6

 Most Attacked Country

India

Highest incident count

 Avg Response Time

90.78 Min

Target: under 60 min

 Total Locations

10

Countries attacked

 Max Response Time

180 Min

Slowest response recorded

Map Visual — Global Threat Hotspot Map

 World map with red bubbles showing attack locations across 10 countries

✓ Why: Map is best for geographic data — instantly shows which country is most attacked

Bar Chart — Avg Response Time by Location

 10 countries ranked by average response time — India highest at 91.3 min

✓ Why: Bar chart compares response efficiency — identifies countries needing improvement

Matrix Table — Industry vs Location Heatmap

 10 countries × 8 industries showing incident count with color coding

✓ Why: Heatmap shows two-dimensional patterns — red = critical, green = low

Key Insights Panel

 3 summary lines — India & USA most attacked, 90 min avg response, Containment most used

✓ Why: Gives executives a quick text summary of the most important geospatial findings



Dashboard Page 4 — Executive Summary

Milestone 4 — Weeks 7–8

⚡ Overall TSI Score

5.21

📄 SLA Compliance

33.3%

✅ Attacks Resolved

~50,030

💾 Total Data at Risk

5.01M GB

🚨 Critical Attacks

12,499

⌚ Avg Attack Duration

151.8 Min



Line Chart — Incident Trend & TSI Score by Month



Two lines — Blue = incident count, Red = TSI score trend over 12 months



Why: Line chart shows change over time — identify upward/downward trends monthly



Table — Security Risk Scorecard



Industry × Severity × TSI Score table — Energy & Finance marked Critical



Why: Scorecard gives quick executive view — which industry is most at risk



Bar Chart — Industry Risk Analysis



8 industries ranked by incident count — Government highest at 12,645



Why: Bar chart compares all industries — helps prioritize security budgets



Gauge — Platform Risk Level



TSI Gauge 5.21 vs Target ≤ 5.0 with SLA 33.4% and Uptime 99.1%



Why: Gauge gives instant risk level — Red zone means action needed



Key Findings



Brute Force is the most common attack

12,605 incidents



Government sector is most targeted industry

12,645 incidents



India has highest avg response time

91.3 minutes



March recorded highest monthly incidents

8,600 incidents



Endpoint Detection is most used security tool

145,231 deployments



50% of attacks were successfully stopped

50,030 resolved



Total data compromised across all incidents

5.01 Million GB



Overall Threat Severity Index score

5.21 — HIGH THREAT



SLA Compliance rate — below 60 min target

Only 33.3%



Ransomware is the only Critical level attack

12,499 incidents



Tools & Technologies Used



Python

📌 Core programming language for data processing

✅ Used for: ETL pipeline, Star Schema creation, TSI calculation



Power BI Desktop

📌 Business intelligence and visualization tool

✅ Used for: 4-page interactive dashboard with navigation and slicers



Jupyter Notebook

📌 Python development environment

✅ Used for: ETL pipeline development and data exploration



Pandas

📌 Data manipulation and transformation library

✅ Used for: Data cleaning, feature engineering, CSV export



Star Schema

📌 Data warehouse modeling technique

✅ Used for: 1 fact table + 5 dimension tables for fast analytics



GitHub

📌 Version control and project hosting

✅ Used for: Repository, README, project documentation



Evaluation Criteria — All Milestones Met

Milestone 1 — Week 2

All datasets integrated successfully ✓

Incident classifications standardized across sources ✓

Star schema relationships validated ✓

🏆 1,00,000 records loaded | 6 tables created | TSI calculated

Milestone 2 — Week 4

Threat Severity Index accurately calculated ✓

Temporal analytics dashboards operational ✓

Anomaly detection functioning correctly ✓

🏆 TSI = 5.21 | Monthly trend built | Anomaly alerts added

Milestone 3 — Week 6

Threat hotspot maps validated ✓

Response performance metrics verified ✓

Geospatial intelligence dashboards fully functional ✓

🏆 World map built | 10-country response bar | Heatmap created

Milestone 4 — Week 8

Risk metrics identify meaningful threat patterns ✓

Predictive models generate actionable forecasts ✓

Executive dashboards support end-to-end decisions ✓

Platform performance meets deployment requirements ✓

🏆 Risk scorecard | Forecast line | Platform gauge | Executive dashboard



Thank You!

Advanced Threat Incident Analytics & Security Visualization Framework

1,00,000

Records Analyzed

6 Tables

Star Schema

4 Pages

Power BI Dashboard

5.21 TSI

Threat Score

Infosys Springboard Virtual Internship 7.0 | Batch 2 | Data Visualization

"Intelligent Security. Actionable Insights. Safer Tomorrow."

GitHub: github.com/mr-deepak-kp/ThreatAnalytics-Dashboard